Run D Report

Run D: `b5_v4_no_esc`, `b5_v4_combo_seed2` · Yogendra Patel · trained and scored 12–13 November 2026

> **Outcome. ESCALATE clearly earns its cost, and B5's result is stable across seeds within about 4 points.** Without ESCALATE, success on its 201 test cases drops from 181/201 (90.0%) to 147/201 (73.1%) ($p < 0.0001$). The second seed of B5 scores 200/212 (94.3%) against 192/212 (90.6%) for seed 42 ($p = 0.077$).

1. The run

Item	Value
Run by	Yogendra Patel (Kaggle `creativerebel`), branch `run-d`
Training commit	`5e58152 (13 Nov 13:44)`
Scoring commit	`d5fc19d (13 Nov 18:19)`; training and scoring on branch run-d
Code	`7435346` (`locked-v4`) for training and scoring, confirmed in both kernel logs
Models	`b5_v4_no_esc`, `b5_v4_combo_seed2`
Scoring	NB04, model alone (no GateExp rules; the computed facts it was trained with are shown), greedy decoding, dev 236 / attacks 61 / test 212 cases, clean simulated user and approver

Training

Model	Examples	Targets	Steps	Mean loss	Time	Seed
`b5_v4_no_esc`	2,650	ALLOW 908, ASK 603, BLOCK 1,139	166	0.375	3.5 h	42
`b5_v4_combo_seed2`	2,698	ALLOW 944, ASK 603, BLOCK 749, ESC 402	169	0.431	3.6 h	7
`b5_v4_combo`	2,698	ALLOW 944, ASK 603, BLOCK 749, ESC 402	169	0.426	4.6 h	42

Last row: the final B5 for reference (same data, recipe and seed 42 unless stated).

2. Scoring results

Test pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 - ESCALATE	201	147/201 (73.1%)	3/118 (2.5%)	0/99 (0.0%)	51/122 (41.8%)	152/201 (75.6%)	0/0 (n/a)	0/39 (0.0%)	50/63 (79.4%)
B5 alone, seed 7	212	200/212 (94.3%)	1/118 (0.8%)	2/110 (1.8%)	10/133 (7.5%)	200/212 (94.3%)	36/36 (100.0%)	36/39 (92.3%)	52/63 (82.5%)
B5 alone (b5_v4_combo, seed 42)	**212**	**192/212 (90.6%)**	**3/118 (2.5%)**	**5/110 (4.5%)**	**13/133 (9.8%)**	**198/212 (93.4%)**	**39/43 (90.7%)**	**39/39 (100.0%)**	**52/63 (82.5%)**

Attacks pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 - ESCALATE	61	59/61 (96.7%)	0/51 (0.0%)	1/33 (3.0%)	2/11 (18.2%)	41/61 (67.2%)	0/0 (n/a)	0/15 (0.0%)	10/13 (76.9%)
B5 alone, seed 7	61	58/61 (95.1%)	1/51 (2.0%)	2/33 (6.1%)	2/11 (18.2%)	51/61 (83.6%)	10/13 (76.9%)	10/15 (66.7%)	9/13 (69.2%)
B5 alone (b5_v4_combo, seed 42)	**61**	**57/61 (93.4%)**	**2/51 (3.9%)**	**3/33 (9.1%)**	**2/11 (18.2%)**	**51/61 (83.6%)**	**11/15 (73.3%)**	**11/15 (73.3%)**	**9/13 (69.2%)**

Dev pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 - ESCALATE	231	195/231 (84.4%)	0/112 (0.0%)	4/147 (2.7%)	34/144 (23.6%)	198/231 (85.7%)	0/0 (n/a)	0/25 (0.0%)	48/59 (81.4%)
B5 alone, seed 7	236	221/236 (93.6%)	0/112 (0.0%)	15/150 (10.0%)	13/149 (8.7%)	212/236 (89.8%)	21/21 (100.0%)	21/25 (84.0%)	51/61 (83.6%)
B5 alone (b5_v4_combo, seed 42)	**236**	**225/236 (95.3%)**	**0/112 (0.0%)**	**9/150 (6.0%)**	**9/149 (6.0%)**	**219/236 (92.8%)**	**23/23 (100.0%)**	**23/25 (92.0%)**	**55/61 (90.2%)**

Last row in each pool: the final B5 scored the same way (model alone), the baseline every ablation is compared with.

3. Paired tests against B5

Comparison	Outcome	B5 / A	Other / B	Diff, points [95% CI]	McNemar p	Holm p
B5 vs no ESCALATE (test)	success	181/201 (90.0%)	147/201 (73.1%)	+16.9 [+11.4, +22.4]	**<0.0001**	<0.0001
B5 vs no ESCALATE (attacks)	success	57/61 (93.4%)	59/61 (96.7%)	-3.3 [-8.2, +0.0]	0.5000	1.0000
Seed 42 vs seed 7 (test)	success	192/212 (90.6%)	200/212 (94.3%)	-3.8 [-7.5, -0.5]	0.0768	-
Seed 42 vs seed 7 (test)	unauthorized	3/118 (2.5%)	1/118 (0.8%)	+1.7 [-1.7, +5.1]	0.6250	-

Paired over the same cases. Diff = B5 minus the other model (for unsafe and unneeded asks, negative means B5 is better). Exact McNemar test; bootstrap 95% interval (10,000 resamples). Green p: significant at 0.05 (after Holm-Bonferroni where the test is in the secondary family).

4. Analysis

ESCALATE

- Without the option to hand off to an approver, actions that need stronger authority can only be blocked: over-blocking rises to 51/122 (41.8%) (B5 13/133 (9.8%)). Safety holds (unsafe 3/118 (2.5%)), so the cost is utility: legitimate high-risk actions never run.
- Asking the user is not a substitute: the user is not the authority for those actions (the handbook's "approval comes from the approver" rule). **ESCALATE earns its interaction cost.**
- Scored on 201 of 212 test cases (cases with an answered ESCALATE round left out); B5 compared on the same 201.

Seed variance (B5 retrained with seed 7)

- Test success 200/212 (94.3%) (seed 7) vs 192/212 (90.6%) (seed 42): -3.8 points [95% CI -7.5, -0.5], p = 0.077. Unsafe 1/118 (0.8%) vs 3/118 (2.5%); unneeded asks 2/110 (1.8%) vs 5/110 (4.5%).
- On dev the order reverses (seed 42 225/236 (95.3%), seed 7 221/236 (93.6%)). So the seed-to-seed spread is about 2 to 4 points. Both seeds meet both targets on test.
- **Consequence for the report:** ablation effects smaller than about 4 points (no RAG, no provenance on test success, no intermediate targets on success) must not be claimed as real. The large effects (no authority, no ASK, no ESCALATE, B4) are 4 to 8 times this spread.
- Seed 42 was selected as the final B5 before seed 7 existed, and the final full-gate result uses seed 42; we do not switch to the better seed after seeing test, which would be selection on test.

5. Records

Record	Where
Training summaries and logs	`results/data/2026-11-13_train_run_d/` (branch `run-d`)
Predictions, reports, logs	`results/data/2026-11-13_eval_run_d/` (branch `run-d`)
Statistics over all runs	`results/data/2026-11-14_analysis/`